AI for Theology

Mentored Research Seminar

Format: 14 weeks · one 100-minute session/week **Type:** Mentored research seminar (students produce an original agentic-AI research project)

Organizing idea: AI as a *scholarly instrument* **and** as a *theological subject* — both "agentic AI **for** studying theology" and "what theology says **about** agentic AI."

Three technical pillars: Agentic AI · Retrieval-Augmented Generation (RAG) · Evaluation.

1. Course description

This seminar teaches students to use **agentic AI** — systems that plan, use tools, retrieve sources, and act over multiple steps — as instruments of theological and religious-studies research, while critically examining the theological and ethical questions autonomous AI raises. Students learn to *design a research process*: an agent that retrieves primary sources (**RAG**), cross-references them, drafts with citations, critiques its own output, and — crucially — is **evaluated** so its claims are checked rather than trusted. Each student (or pair) develops, executes, evaluates, and defends an original project.

The three pillars build on each other: **agentic AI** is the system, **RAG** is how it stays grounded in real texts, and **evaluation** is how you know whether any of it is trustworthy — especially vital where a fabricated scripture reference is a scholarly (and pastoral) failure.

2. Learning objectives

By the end of the course a student can:

1. **[Agentic AI]** Explain what turns an LLM into an agent (tool use, memory, plan→act→observe, autonomy) and build a multi-step, self-critiquing research agent.
2. **[RAG]** Build a retrieval-augmented pipeline over a theological corpus so the agent cites *real* sources, and diagnose where grounding fails.
3. **[Evaluation]** Design and run an evaluation of a RAG/agent system — retrieval quality, groundedness/faithfulness, answer relevance, task success, and cost — using golden datasets, LLM-as-judge, and human review, while knowing each method's limits.
4. Design a researchable theological question with a matching agentic method **and eval plan**.
5. Analyze the theological questions autonomy raises — personhood, *imago Dei*, agency, responsibility, spiritual authority — and report AI-assisted scholarship **honestly**.

3. Track decision — Hybrid (low-code default, optional light-code)

Low-code by default: students build with provided notebook templates and a config-driven builder, keeping the focus on theology and research design. A parallel **light-code (Python) path** offers the same milestones for technically inclined students. Every lab ships with a working scaffold — no one writes agents or eval harnesses from scratch.

4. Standardized toolkit

Layer	Default (low-code)	Light-code option
Model access	Hosted chat model with tool-use (instructor-provided)	Same, via API
Agent scaffold	Notebook templates ("research agent," "researcher + fact-checker")	Minimal Python agent loop
Retrieval (RAG)	Prebuilt vector-search over the class corpus	Student-assembled components
Evaluation	Provided eval notebook (golden Q&A set, groundedness + citation checks, LLM-as-judge template)	Same, extensible (e.g. a RAG-eval library)
Corpus/sources	Curated digital primary sources (see Week 3)	Same
Documentation	Shared trace log + prompt/tool/eval journal	Same

Guest session recommended around Week 5–6: "How real agentic systems are built *and evaluated*" from a practitioner.

5. Weekly session shape (100 min)

∼40 min concept & discussion · ∼40 min hands-on lab · ∼20 min project touchpoint (peer review or mentor check-in). Continuous mentoring, not just in Phase 4.

Phase 1 — Foundations (Weeks 1–3)

Week 1 Framing: "AI for Theology"

- **Objective:** Locate the field; articulate a personal research interest.
- **Topics:** Instrument vs. subject; digital humanities, computational text study, AI ethics; survey of the AI/religion intersection; preview the three pillars.
- **Hands-on:** Map "questions I care about"; a deliberately naive chatbot query on a doctrine question — note what feels off (previews *why* we need RAG + evaluation).
- **Deliverable:** One-paragraph interest statement.
- **Readings:** A survey on AI & religion (e.g. Beth Singler); course intro notes.

Week 2 From LLMs to *agents* **[Agentic AI]**

- **Objective:** Build the core mental model of an agent.

- **Topics:** Next-token prediction, embeddings, hallucination; the step to agency — tool use, memory, the plan→act→observe loop, autonomy.
- **Hands-on:** Watch an agent decompose a question and call a tool; diagnose where it errs.
- **Readings:** Wolfram, *What Is ChatGPT Doing . . . ?* (selections); Anthropic, "Building Effective Agents" (2024); Weng, "LLM-Powered Autonomous Agents" (2023). *Optional:* Yao et al., "ReAct."

Week 3 Sources and the theological corpus

- **Objective:** Understand the data landscape and its theological freight.
- **Topics:** Digital primary sources (original-language scripture, patristics, Qur'anic corpora, Talmud, translations, hymnody, sermons); copyright, canon, translation politics; "the data is theologically loaded" — and why that shapes both retrieval and evaluation.
- **Hands-on:** Locate and load the corpus each student will use all term.
- **Readings:** Claire Clivaz on digital humanities & biblical studies (selection).

Phase 2 — Technical Core: Agentic AI · RAG · Evaluation (Weeks 4–7)

Week 4 Retrieval-Augmented Generation **[RAG]**

- **Objective:** Make an agent cite *real* texts, not invented ones.
- **Topics:** Embeddings and semantic search; chunking; the retrieve→augment→generate loop; RAG as the agent's key tool; failure modes (bad chunking, wrong retrieval, ignored context).
- **Hands-on:** Build an "ask-your-corpus" RAG agent over a Week-3 text.
- **Readings:** Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks" (2020, the idea, not the math).

Week 5 Building a research agent **[Agentic AI]**

- **Objective:** Turn a RAG tool into a multi-step, self-critiquing research agent.
- **Topics:** Agent design patterns — planning, reflection/self-critique; a two-agent setup ("researcher" + "critic/fact-checker") to catch fabricated references; text-analysis techniques (semantic search, concept-tracing of *hesed*/*agape*/*grace*) reframed as **tools the agent wields**.
- **Hands-on:** Extend Week-4's RAG into a planning + reflection agent; add a fact-checker.
- **Readings:** Shinn et al., "Reflexion" (2023, concept); Wang et al., "A Survey on LLM-based Autonomous Agents" (2023, selections). *Guest slot.*

Week 6 Evaluation: how do you know it works? **[Evaluation]** ★ new

- **Objective:** Measure whether a RAG/agent system is trustworthy.
- **Topics:**
  - **RAG evaluation** — retrieval precision/recall, **groundedness/faithfulness** (is the answer supported by retrieved text?), answer relevance, and **citation verification** (does the cited verse/passage actually exist and say that?).

  - **Agent evaluation** — task success, trajectory / tool-use correctness, cost & latency efficiency, robustness.
  - **Methods** — golden/reference datasets, **LLM-as-judge** and its pitfalls (bias, inconsistency), human evaluation, and when to use which.
  - **Theology-specific angles** — doctrinal fidelity, translation sensitivity, and **bias across traditions** (whose theology is the "correct answer"?).
- **Hands-on:** Build a small golden Q&A set for your corpus; run groundedness + citation-existence checks and an LLM-as-judge pass on Week-5's agent; report a scorecard.
- **Readings:** Es et al., "RAGAS" (2023); Liu et al., "G-Eval" (2023); Zheng et al., "Judging LLM-as-a-Judge / MT-Bench" (2023, selections).

Week 7 Research design & proposal (midpoint gate)

- **Objective:** Convert curiosity into a researchable question + agent design + **eval plan**.
- **Topics:** Scope, method, evidence, what counts as a finding; which agent steps and tools; how you will **evaluate** and verify claims (metrics, golden set, human check); reproducibility and documenting agentic-AI use (prompts, tools, traces, eval).
- **Deliverable:** 1–2 page proposal + 5-min pitch. **Must include an evaluation plan.**
- **Readings:** Research-methods primer (notes); one exemplar agentic-DH study.

Phase 3 — Domain Deep-Dives + Project Launch (Weeks 8–10)

Week 8 AI and hermeneutics

- **Objective:** Interrogate whether an agent can "interpret."
- **Topics:** Multi-step reasoning vs. genuine understanding; authorial intent, tradition, the interpretive community; higher stakes when an agent *acts* on interpretation.
- **Hands-on:** Projects formally begin; first evaluated agent runs on the student's question.
- **Readings:** Gadamer, *Truth and Method* (selections); a piece on machine "understanding."

Week 9 Theology *of* agentic AI

- **Objective:** Examine personhood and autonomy theologically.
- **Topics:** *Imago Dei*, personhood, consciousness; **agency and autonomy** — moral status, responsibility, delegation when a system *acts*; AI and idolatry; eschatology.
- **Hands-on:** Mentored project check-in.
- **Readings:** Herzfeld, *The Artifice of Intelligence* (2023) selections; Dorobantu on AI & personhood; Vatican, *Rome Call for AI Ethics* (2020) / *Antiqua et Nova* (2025) selections.

Week 10 Ethics, bias, and the limits of autonomy

- **Objective:** Adopt a critical, human-in-the-loop stance (connects back to evaluation).

- **Topics:** Training-data bias (whose theology dominates?); fabricated citations; agentic-specific risks — compounding errors across steps, over-trust in an autonomous "researcher," pastoral/spiritual-authority harms; designing human oversight; **bias-aware evaluation**.
- **Hands-on:** Each project adds an explicit verification/oversight step to its eval plan.
- **Readings:** Bender et al., "On the Dangers of Stochastic Parrots" (2021); Crawford, *Atlas of AI* (2021, selections).

Phase 4 — Mentored Research Execution (Weeks 11–13)

Week 11 Build sprint (studio)

- **Objective:** Execute the agentic pipeline; troubleshoot traces.
- **Format:** Studio — students build, mentors circulate; structured trace debugging.

Week 12 Evaluate & critique **[Evaluation, applied]**

- **Objective:** Run a real evaluation of your *own* project and defend it.
- **Format:** Students apply Week-6 methods to their systems (scorecard: groundedness, citation checks, task success, bias spot-checks); peer-review swap + mentor stress-test of claims and AI-reliance.
- **Deliverable:** Preliminary results + **evaluation report** (early-warning checkpoint).

Week 13 Write-up & communication

- **Objective:** Turn results into an honest argument.
- **Topics:** Presenting agentic work with integrity — goals, tools, traces, **eval results**, failures, human oversight. Draft final paper + slides.

Week 14 — Symposium

Final presentations (cohort + invited faculty), Q&A, and a closing reflection: **what did the agent reveal, what did it distort, how well did it hold up under evaluation, and what did this teach us theologically?**

- **Deliverable:** Final paper + presentation.

6. Assessment

Component	Weight	Due
Participation & project touchpoints	15%	Weekly
Interest statement	5%	Week 1
Research proposal + pitch (incl. eval plan)	20%	Week 7 (gate)
Preliminary results + evaluation report	15%	Week 12
Final paper	30%	Week 14
Final presentation	15%	Week 14

7. Rubrics

Research proposal (Week 7)

- **Question (20%)** — theologically substantive, researchable, well-scoped.
- **Agent design (20%)** — steps, tools, and RAG grounding fit the question.
- **Evaluation plan (25%)** — credible metrics + golden set / human check; awareness of failure modes and bias.
- **Feasibility & sources (20%)** — corpus available; scope achievable.
- **Communication (15%)** — clear pitch and writing.

Final paper (Week 14)

- **Scholarly contribution (25%)** — a real finding or a well-argued negative result.
- **Method & agent design (20%)** — sound, well-documented, reproducible RAG/agent pipeline.
- **Evaluation & critical stance (25%)** — claims measured *and* checked; groundedness/citations verified; bias and limits addressed; human-in-the-loop evident.
- **Theological/ethical reflection (15%)** — engages the “AI as subject” questions (Weeks 8–10).
- **Communication (15%)** — structure, clarity, honest reporting of AI use.

Agentic-AI use disclosure (required in every deliverable). An appendix in every submission: models/tools used, key prompts, representative **traces**, **evaluation results/scorecard**, where the agent failed, and how the student verified results. Undisclosed or unevaluated AI reliance is penalized.

8. Notes for the instructor

- **Readings are starting points from the instructor’s knowledge — verify current editions, availability, and access before distributing.** Swap in tradition-appropriate sources (Christian / Jewish / Islamic / comparative) to match the cohort.

- **The three pillars are cumulative** — RAG (W4) grounds the agent (W5), and evaluation (W6) judges both; keep labs building on the *same* corpus and agent so students see the through-line.
- **Evaluation is the highest-leverage theology-specific skill here** — a fabricated citation is a scholarly failure, so budget real time for the golden-set + citation-verification lab.
- **Decide access early:** hosted model + tool-use quota; whether the light-code path needs an optional extra lab.
- **Corpus curation is the top prep task** — a clean, rights-clear, well-scoped corpus makes every RAG and eval lab work.